INCH-POUND

MIL-PRF-19500/735
4 October 2006

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, SCHOTTKY, DUAL, CENTER TAP,
TYPE 1N7041CCU1 AND SINGLE DIODE TYPE 1N7045T3, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of
this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, schottky, power rectifier diodes for use in high frequency switching applications. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (U-1) and figure 2 (TO-257AA).

1.3 Maximum ratings. Unless otherwise specified, maximum ratings (T_C = +25°C).

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6
Types	V_{RWM}	I_O (1) T_C = +100°C	I_{FSM} (2) t_p = 8.3 ms, T_C = +25°C	$R_{\Theta JC}$ (2)(3)	T_{STG} and T_J
	V dc	A dc	A (pk)	°C/W	°C
1N7041CCU1	45	20 (2)	250 (2)	1.67	-65 to +150
1N7045T3	45	10	110	2.6	

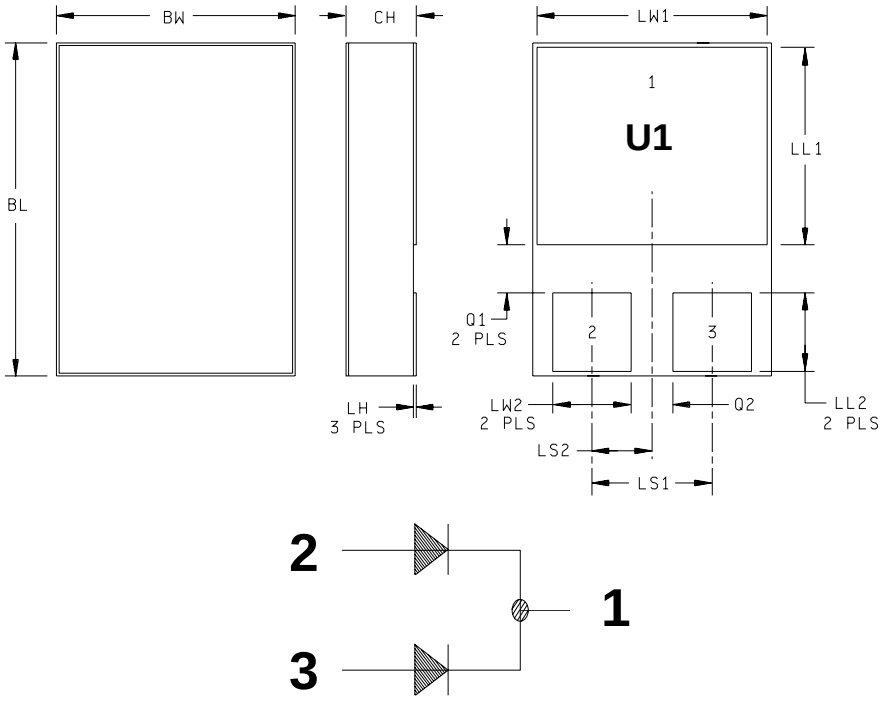
(1) See temperature-current derating curves in figures 3 and 4.
(2) Each leg.
(3) See figures 5 and 6.

1.4 Primary electrical characteristics. $R_{\Theta JC}$ = 0.83°C/W maximum for entire package for 1N7041CCU1; $R_{\Theta JA}$ = 40°C/W maximum for each leg; $R_{\Theta JC}$ = 2.6°C/W maximum for 1N7045T3.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to semiconductor@dscc.dla.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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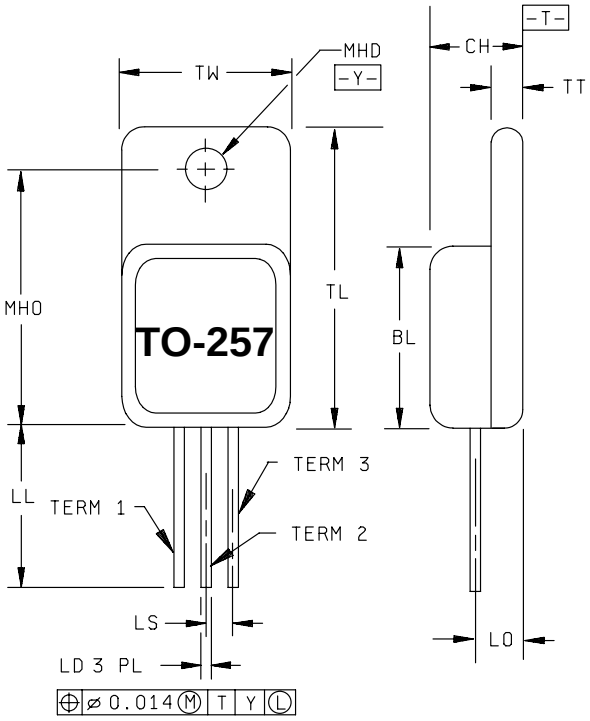


Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.620	.630	15.75	16.00
BW	.445	.455	11.30	11.56
CH	.129	.139	3.28	3.53
LH	.010	.020	0.25	0.51
LL1	.410	.420	10.41	10.67
LL2	.152	.162	3.86	4.11
LS1	.200	.220	5.08	5.59
LS2	.100	.110	2.54	2.79
LW1	.370	.380	9.40	9.65
LW2	.135	.145	3.43	3.68
Q1	.035		0.89	
Q2	.050		1.27	
Term 1	Cathode			
Term 2	Anode			
Term 3	Anode			

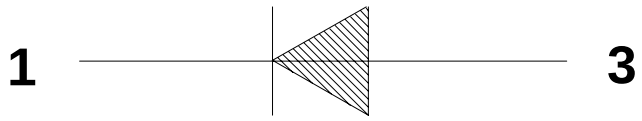
NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 1. Physical dimensions and configuration for 1N7041CCU1.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.410	.430	10.41	10.92
CH	.190	.200	4.83	5.08
LD	.025	.040	0.64	1.02
LL	.500	.750	12.70	19.05
LO	.120 BSC		3.05 BSC	
LS	.100 BSC		2.54 BSC	
MHD	.140	.150	3.56	3.81
MHO	.527	.537	13.39	13.64
TL	.645	.665	16.38	16.89
TT	.035	.045	0.89	1.14
TW	.410	.420	10.41	10.67



NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 2. Dimensions and configuration for 1N7045T3.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 and 2 herein. Methods used for electrical isolation of the terminal feedthroughs for the TO-257 package shall employ materials that contain a minimum of 90 percent Al_2O_3 (ceramic).

3.4.1 Polarity. Polarity and terminal configuration shall be in accordance with figures 1 and 2 herein.

3.4.2 Lead finish and formation. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of finish is desired, it shall be specified in the acquisition document (see 6.2). When lead formation is performed on the TO-257, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14 of MIL-PRF-19500 and 100 percent dc testing in accordance with table I, subgroup 2 herein.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in tables I and II herein.

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3.7 Marking. Marking shall be in accordance with MIL-PRF-19500 and herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II herein).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (JANS, JANTXV, and JANTX levels). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
(1) (2) 3b	Method 4066 of MIL-STD-750, Condition A, one pulse, I_O = 0, V_{RWM} = 0, see 1.3 herein, column 4.	Method 4066 of MIL-STD-750, Condition A, one pulse, I_O = 0, V_{RWM} = 0, see 1.3 herein, column 4.
(1) 3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
3d	Peak reverse energy test (see 4.3.3)	Peak reverse energy test (see 4.3.3)
4	Required	Optional
5	Required	Not applicable
8	Required	Not applicable
9, 10	Not applicable	Not applicable
11	V_{F1} and I_{R1}	V_{F1} and I_{R1}
12	See 4.3.1	See 4.3.1
13	Subgroup 2 and 3, of table I herein, V_{F1} and I_{R1}, excluding thermal impedance; ΔV_{F1} ≤50 mV (pk); ΔI_{R1} = ± 100 percent from the initial value or 100 uA dc, whichever is greater.	Subgroup 2, of table I herein excluding thermal impedance; V_{F1} and I_{R1}; ΔV_{F1} ≤50 mV (pk); ΔI_{R1} = ± 100 percent from the initial value or 100 uA dc, whichever is greater.
15	Required	Not applicable
16	Required	Not applicable

(1) Thermal impedance and surge shall be performed any time after screen 3a and before screen 13.
(2) Surge shall precede thermal impedance.

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4.3.1 Power burn-in conditions. Burn-in conditions are as follows: Method 1038 of MIL-STD-750, test condition A. T_C = +125°C; V_R = 36 V dc.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} and V_C. Measurement delay time (t_{MD}) = 70 μs max. See table III, group E, subgroup 4 herein.

4.3.3 Peak reverse energy test. The peak reverse energy test is to be performed using the circuit as shown on figure 7 or equivalent. The Schottky rectifier under test must be capable of absorbing the reverse energy, as follows: I_{RM} = 1 A, V_{RSM} = 45 V minimum, L = 100 uH.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500, and table I herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables E-VIa (JANS) and E-VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2, forward voltage test (V_{F1}) and reverse leakage test (I_{R1}) herein. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Group B inspection, table E-VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	ΔT_C = +85°C, I_F = 2 A minimum for 2,000 cycles.
B5	1038	Condition B, I_F = 1 A dc minimum, adjust T_A and I_F to achieve T_J = +150°C, +0° -10°C, t = 240 hours minimum; (heat sinking allowed).
B6	4081	Limit for thermal resistance for 1N7041 is 1.67°C/W for each leg. Limit for thermal resistance for 1N7045 is 2.6°C/W.

4.4.2.2 Group B inspection, table E-VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1037	ΔT_C = +85°C minimum, I_F = 2 A minimum for 2,000 cycles.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2, forward voltage test (V_{F1}) and reverse leakage test (I_{R1}) herein. Delta measurements shall be in accordance with table II herein.

Subgroup	Method	Condition
C2	2036	Condition A, weight = 10 lbs, t = 15 seconds.
C6	1037	ΔT_C = +85°C, minimum, I_F = 2 A minimum for 6,000 cycles.

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4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-IX of MIL-PRF-19500, and table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein except $Z_{\Theta JX}$ need not be performed after group E subgroup 2.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.3.2	$Z_{\Theta JX}$			
1N7041					1.35	ºC/W
1N7045					1.60	ºC/W
Forward voltage	4011	Pulsed (see 4.5.1)	V_{F1}			
1N7041		I_F = 10A (pk)			0.57	V dc
1N7045		I_F = 10A (pk)			0.73	V dc
Forward voltage	4011	Pulsed (see 4.5.1)	V_{F2}			
1N7041		I_F = 20 A(pk)			0.69	V dc
1N7045		I_F = 20 A(pk)			1.02	V dc
Reverse current	4016	DC method	I_{R1}			
1N7041		V_R = 45 V			0.5	mA dc
1N7045		V_R = 45 V			0.5	mA dc
Subgroup 3						
High temperature operation:		T_C = +125 °C				
Forward voltage	4011	Pulsed (see 4.5.1)	V_{F3}			
1N7041		I_F = 10 A(pk)			0.48	V dc
1N7045		I_F = 10 A(pk)			0.70	V dc
Forward voltage	4011	Pulsed (see 4.5.1)	V_{F4}			
1N7041		I_F = 20 A(pk)			0.61	V dc
1N7045		I_F = 20 A(pk)			1.10	V dc
Reverse current	4016	DC method;	I_{R2}			
1N7041		V_R = 45 V			15	mA dc
1N7045		V_R = 45 V			67	mA dc

See footnotes at end of table.

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TABLE I. Group A inspection – Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - continued						
Low temperature operation:		T_C = -55°C				
Forward voltage	4011	Pulsed (see 4.5.1)	V_{F5}			
1N7041		I_F = 10 A dc			0.62	V dc
1N7045		I_F = 10 A dc			0.74	V dc
Forward voltage	4011	Pulsed (see 4.5.1)	V_{F6}			
1N7041		I_F = 20 A dc			0.67	V dc
1N7045		I_F = 20 A dc			0.95	V dc
Subgroup 4						
Junction capacitance	4001	V_R = 5 V dc, f = 1 MHz, V_{SIG} = 50 mV (p-p)	C_J		900	pF
Subgroup 5						
Not applicable						
Subgroup 6						
Surge	4066	See column 4 of 1.3. Ten surges for each internal diode. 60 seconds between surges. (see 4.5.1)				
Electrical measurements		See table I, subgroup 2 herein.				
Subgroup 7						
Dielectric withstanding voltage	1016	V_R = 500 V dc; all leads shorted, measure from leads to case	DWV		10	μA dc
Scope display evaluation	4023	Stable only				
Electrical measurements		See table I, subgroup 2 herein.				

1/ For sampling plan, see MIL-PRF-19500.
2/ Electrical characteristics apply to all package styles and polarities.

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TABLE II. Groups B, C, and E delta requirements. 1/ 2/ 3/ 4/ 5/ 6/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage 1N7041 1N7045	4011	 I_F = 10 A (pk) pulsed (see 4.5.1) I_F = 10 A (pk) pulsed (see 4.5.1)	ΔV_{F1}	±50 mV dc from initial reading.		
2.	Reverse current	4016	Vr = 45V	ΔI_{R1}	±100 percent from the initial value or 100μA dc, whichever is greater.		
3.	Thermal impedance	3101	See 4.3.2	$Z_{\Theta JX}$			

1/ Each internal diode.
2/ The electrical measurements for table E-VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 4, see table II herein, steps 1, 2, and 3.
   b. Subgroup 5, see table II herein, steps 1 and 2.
3/ The electrical measurements for table E-VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table II herein, steps 1, 2, and 3.
   b. Subgroup 3, see table II herein, steps 1, 2, and 3.
   c. Subgroup 6, see table II herein, steps 1 and 2.
4/ The electrical measurements for table E-VII of MIL-PRF-19500 are as follows:
   a. Subgroups 2 and 3, see table II herein, steps 1, 2, and 3 for all levels.
   b. Subgroup 6, see table II herein, steps 1, 2, and 3 for all levels.
5/ Devices which exceed the table I limits for this test shall not be accepted.
6/ The electrical measurements for table E-IX of MIL-PRF-19500 are as follows:
   a. Subgroup 1, see table II herein, steps 1, 2, and 3.
   b. Subgroup 2, see table II herein, steps 1 and 2.

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TABLE III. Group E inspection (all quality levels) – for qualification and requalification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			n = 12, c = 0
Temperature cycling (air to air)	1051	Test condition G, 500 cycles, -55°C to +150°C.	
Hermetic seal	1071		
Electrical measurements		See table I, subgroup 2.	
Subgroup 2			n = 12, c = 0
Life test	1048	t = 1,000 hours, T_C = +125°C, V_R = 80 percent rated voltage (see 1.3, column 2 herein).	
Electrical measurements		See table I subgroup 2.	
Subgroup 4			
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 6			n = 3, c = 0
ESD	1020		
Subgroup 7 1/			n = 5, c = 0
Surge	4066		
1N7041		Condition A, T_A = +25°C I_{FSM} = 250 A, ten surges of 8.3 ms superimposed on I_O. V_R = 0; I_O = 10 A pk half sine wave, continuous.	
1N7045		Condition A, T_A = +25°C I_{FSM} = 110 A, ten surges of 8.3 ms superimposed on I_O. V_R = 0; I_O = 10 A pk half sine wave, continuous.	
Electrical measurements		See table I subgroup 2 (V_F and I_R only).	

1/ Each individual diode.

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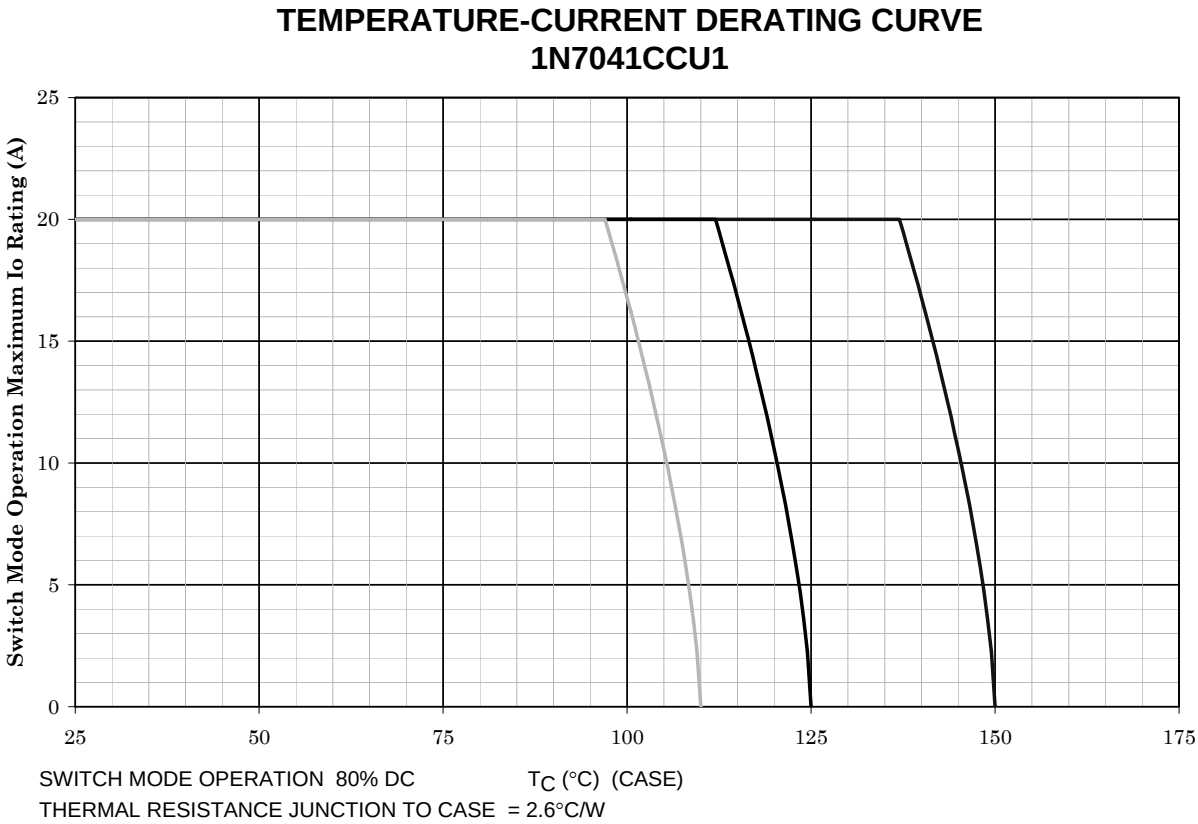


FIGURE 3. Temperature-current derating curve (per leg) 1N7041CCU1.

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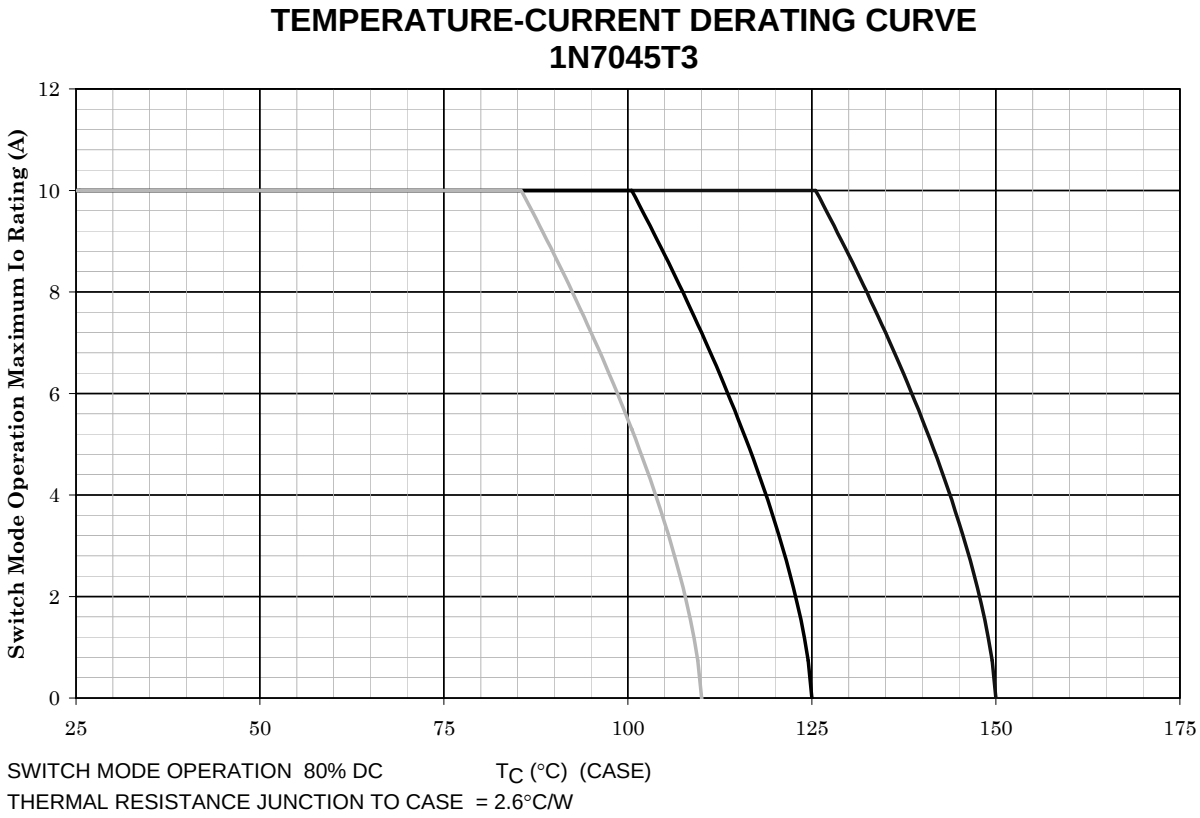


FIGURE 4. Temperature-current derating curve for 1N7045T3.

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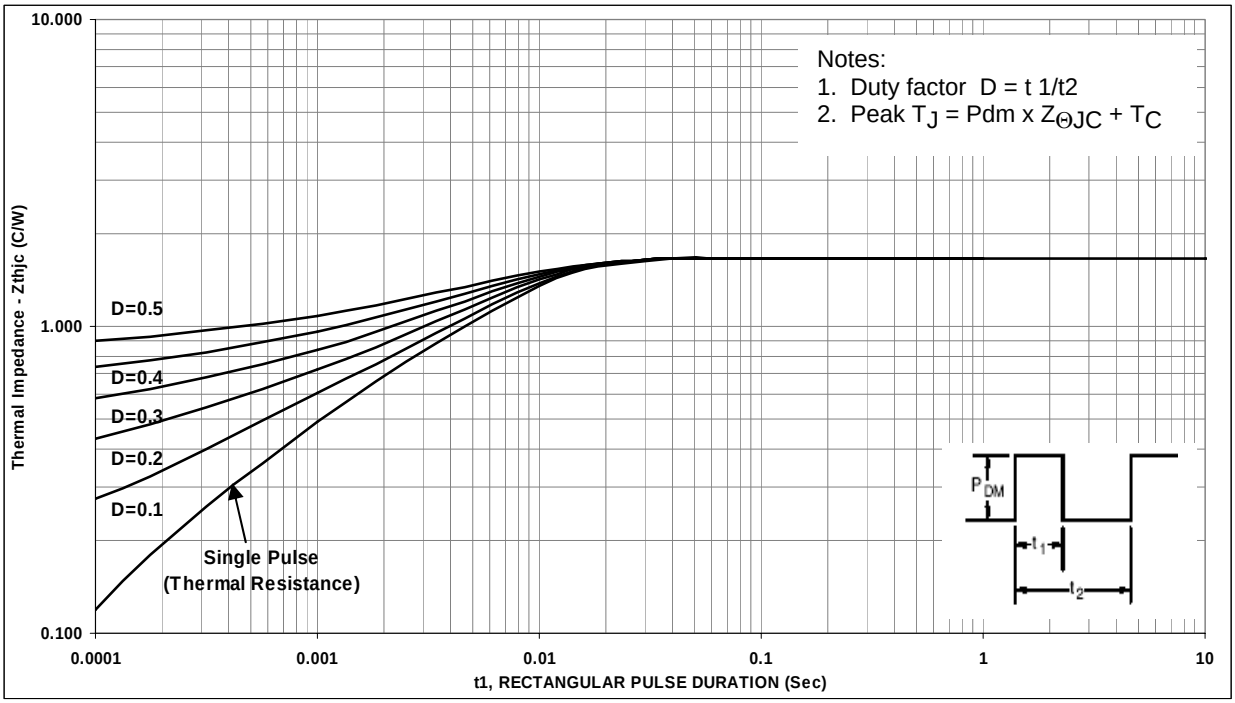


FIGURE 5. Thermal impedance (per leg) 1N7041CCU1.

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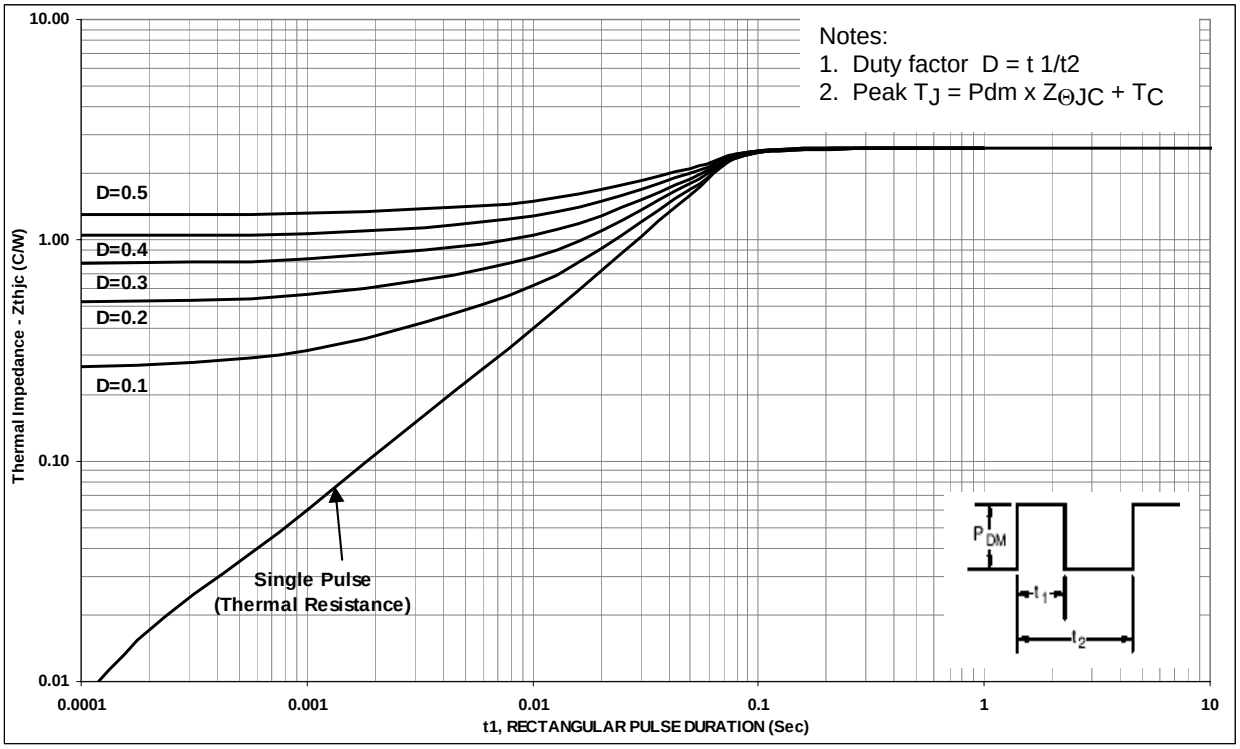
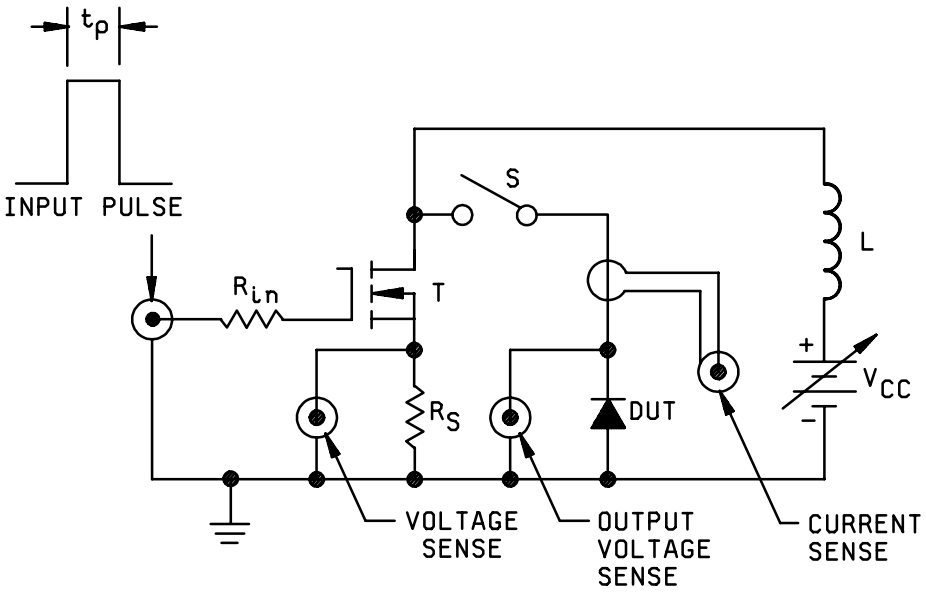


FIGURE 6. Thermal impedance for 1N7045T3.

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Input pulse R_{in} = 50 ohms, 1 watt
V_G = 10 volts, R_S = 0.1 ohms, 1 watt
Z_G = 50 ohms
L = 100uH
P.W. ≈ 30 μs
Duty cycle ≤ 1 percent, T = IRF150/2N6764 or equivalent

Procedures:

1. With S open, adjust pulse width to test current of 1 amps across R_S.
2. Close S, verify test current with current sense.
3. Read peak output voltage (see 4.3.3).

FIGURE 7. Peak reverse energy test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish and formation (see 3.4.2).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Cross reference substitution list. A PIN for PIN replacement table follows, and these devices are directly interchangeable.

Non-preferred PIN	Preferred PIN
20CLQ045 10YQ045C	JANS, JANTXV, JANTX, JAN1N7041CCU1 JANS, JANTXV, JANTX, JAN1N7045T3

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2005-050)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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Check the source to verify that this is the current version before use.